

Technical Note: Translating MSM Forecasts to a Hexagonal Grid

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DRAFT

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Disclaimer

1. Auckland Council has engaged Invisio to prepare the City Rail Link Benefits Realisation Plan (BRP) and undertake Monte Carlo simulations on demand forecasts. Invisio has subcontracted UrbanUmber to deliver the BRP demand forecasting components, including the Monte Carlo simulation work.
2. The demand forecasts, sociodemographic inputs, and public transport network and service inputs used in this analysis are derived from the City Rail Link BRP datasets produced by the Auckland Forecasting Centre (AFC). These underlying datasets are subject to revision as AFC updates its models, assumptions, or source files.
3. This technical note describes (i) a spatial allocation process that redistributes MSM zone-level forecasts to H3 hexagons, and (ii) a process for converting MSM public transport network and service inputs into General Transit Feed Specification (GTFS) format for accessibility modelling. Both outputs are derived products and are dependent on the quality, completeness, and spatial accuracy of the source datasets, including AFC model outputs and publicly available Unitary Plan layers.
4. UrbanUmber has applied reasonable care and professional skill in preparing these derived datasets. However, no warranty is given as to the accuracy, completeness, or suitability of the allocated hexagon-level outputs or GTFS-formatted public transport networks for any specific purpose. Users are responsible for assessing whether the information is appropriate for their intended application.
5. UrbanUmber accepts no liability for any loss, damage, or costs arising from the use or interpretation of the allocated demand, sociodemographic information, or GTFS-formatted public transport network and service data, whether directly or indirectly.

1. Purpose

This Technical Note sets out the rationale and methodology for allocating travel demand forecasts and socio-economic inputs produced at the MSM zone level to H3 hexagons at resolutions 9 and 10. It also describes the process used to convert MSM public transport network and service inputs into General Transit Feed Specification (GTFS) format for use in *r5r* multimodal routing. Together, these workflows enable high-resolution spatial analysis and accessibility modelling on a consistent hexagonal grid.

2. Problem Statement

The primary tools used to develop patronage forecasts in Auckland are land-use models and the regional Macro Strategic travel demand Model (MSM). The Auckland Forecasting Centre (AFC) is mandated by its partner organisations (Auckland Council, Auckland Transport, and Waka Kotahi NZTA) to manage, develop, maintain, and operate Auckland's regional travel demand and traffic models to provide consistent decision support across all major transport projects and policies.

MSM is the endorsed baseline demand model for all major transport projects in Auckland. It has been extensively peer-reviewed for infrastructure investments, multimodal service planning, and policy applications — most recently by Transport for London for a major rapid transit project, with the review concluding that MSM meets international standards and is suitable for assessing rapid transit investments.

However, MSM's spatial resolution is constrained by computational requirements. With 603 zones, and model run-time increasing approximately with the square of the number of zones, the system necessarily relies on relatively large, irregularly shaped census-based zones. These zones often include substantial areas where no development or activity can occur - such as motorways, water bodies, parks, and reserves - and their irregular geometries make spatial visualisation difficult and can obscure meaningful local variation. While the forecasts themselves are robust, the zone structure limits the ability to map, analyse, and communicate forecast spatial patterns of demand with precision.

Allocating MSM forecasts to a regular grid of uber-hexagons addresses these limitations. Hexagonal tessellations are widely recognised in spatial analysis for their uniform shape, equal area, and reduced edge effects compared with irregular polygons. By redistributing MSM demand to hexagons using developable land and other spatial constraints, the method preserves the integrity of MSM forecasts while enabling far more accurate and interpretable spatial representation.

This approach also unlocks additional analytical capability. Converting MSM's public transport networks into GTFS format allows tools such as *r5r* to perform high-resolution accessibility analysis directly on the hexagonal grid. Together, these enhancements provide a more spatially faithful, analytically powerful, and communicatively effective way to work with MSM outputs.

3. Methodology to allocate MSM forecast demands to H3 hexagons

The process assigns MSM forecast demand to H3 hexagons by aligning the hexagon grid to the MSM model area, removing non-developable land, determining each hexagon's share of developable area within its MSM zone, and allocating demand accordingly. The resulting hexagon layer provides a spatially accurate, fully reconciled representation of MSM forecasts suitable for mapping, analysis, and accessibility modelling.

This is shown in Figure 1 and described below with a more detailed explanation in Appendix A:

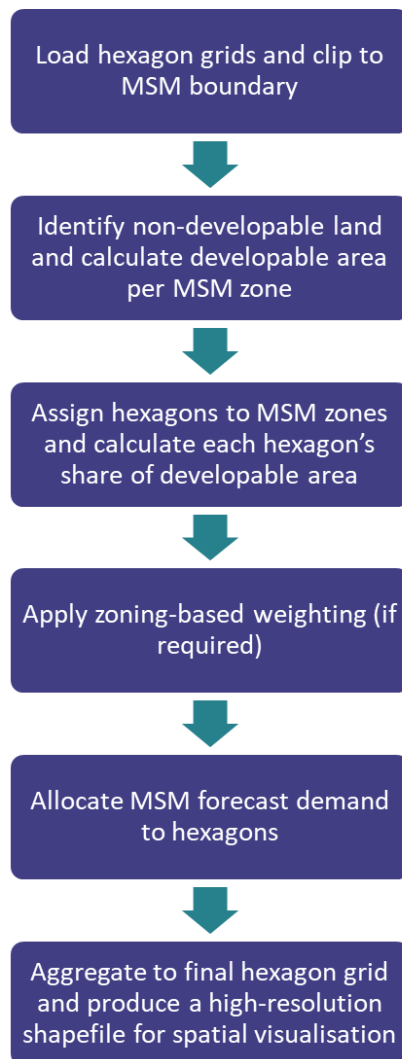


Figure 1: Methodology for Translating MSM Forecasts to a Hexagonal Grid

The overall spatial workflow to allocate MSM forecast demands to H3 hexagons is as follows:

1. Obtain hexagon grids

- Download H3 resolution-9 and resolution-10 hexagons covering the full spatial extent of the MSM model.

2. Align hexagons to the MSM model boundary

- Clip the hexagon grids to the MSM model boundary to match coastlines and the outer extent of the Auckland region.

3. Identify non-developable land

- Create a composite shapefile of areas that cannot be developed using open-source Unitary Plan layers¹ (e.g., road corridors, aerodrome runways, reserves, school grounds, inland water bodies).
- Figure 2 shows the composite developable-land layer (orange) with MSM zone boundaries overlaid in blue.

¹ [Unitary Plan \(Shapefile\) | Auckland Council Open Data](#)

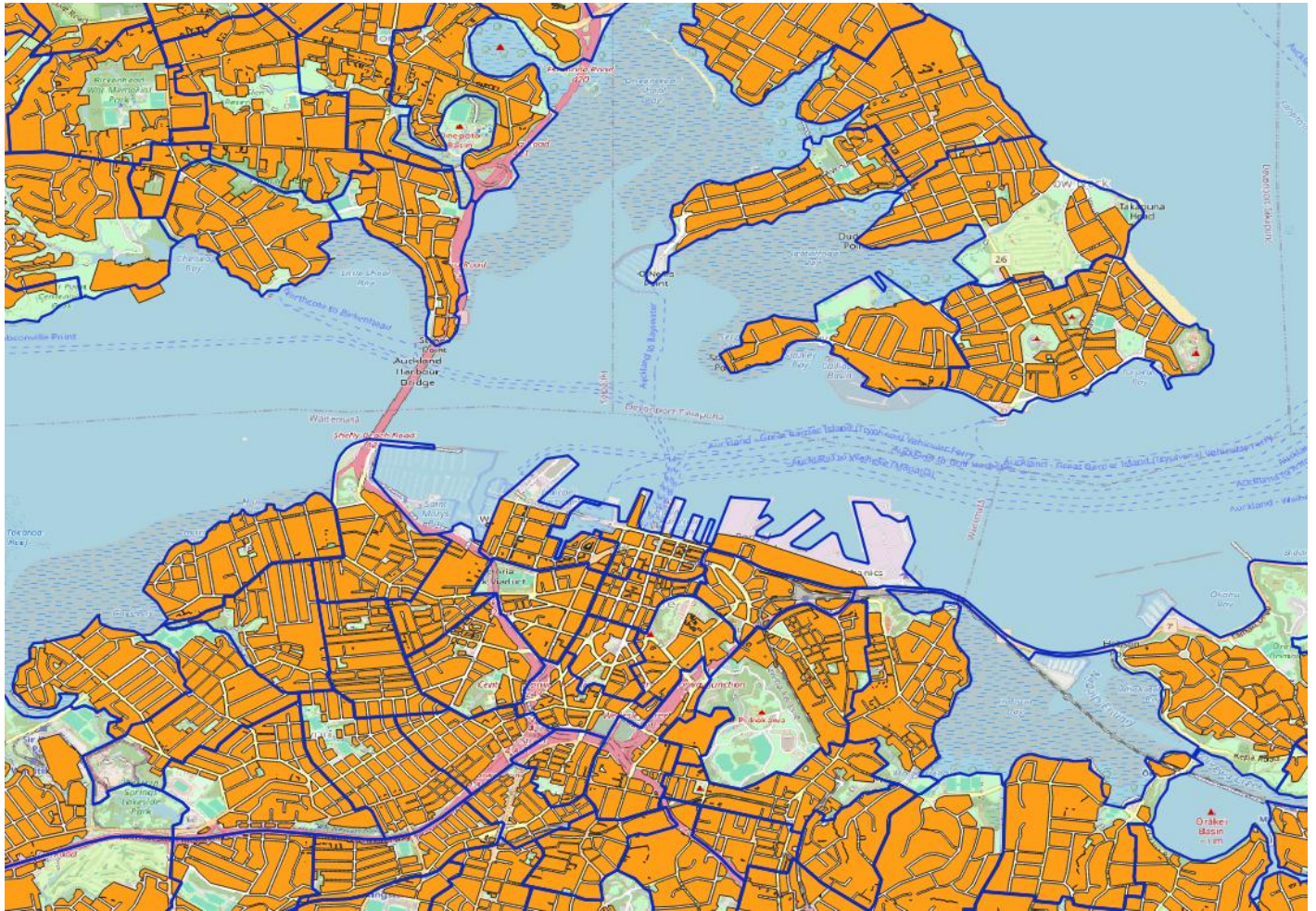


Figure 2: Developable land and MSM Zone boundaries

4. Prepare MSM zone geometry

- Download the MSM zone shapefile from AFC and calculate the total area of each MSM zone.

5. Calculate developable area within each MSM zone

- Overlay the non-developable land shapefile onto the MSM zones.
- Compute the developable area for each MSM zone after removing non-developable land.

6. Assign hexagons to MSM zones

- Spatially join hexagons to MSM zones.
- Where MSM boundaries intersect hexagons, split the hexagons and calculate the area of each resulting piece.
- Figure 3 shows the spatially joined H3 hexagons (resolution 10) and MSM zones.

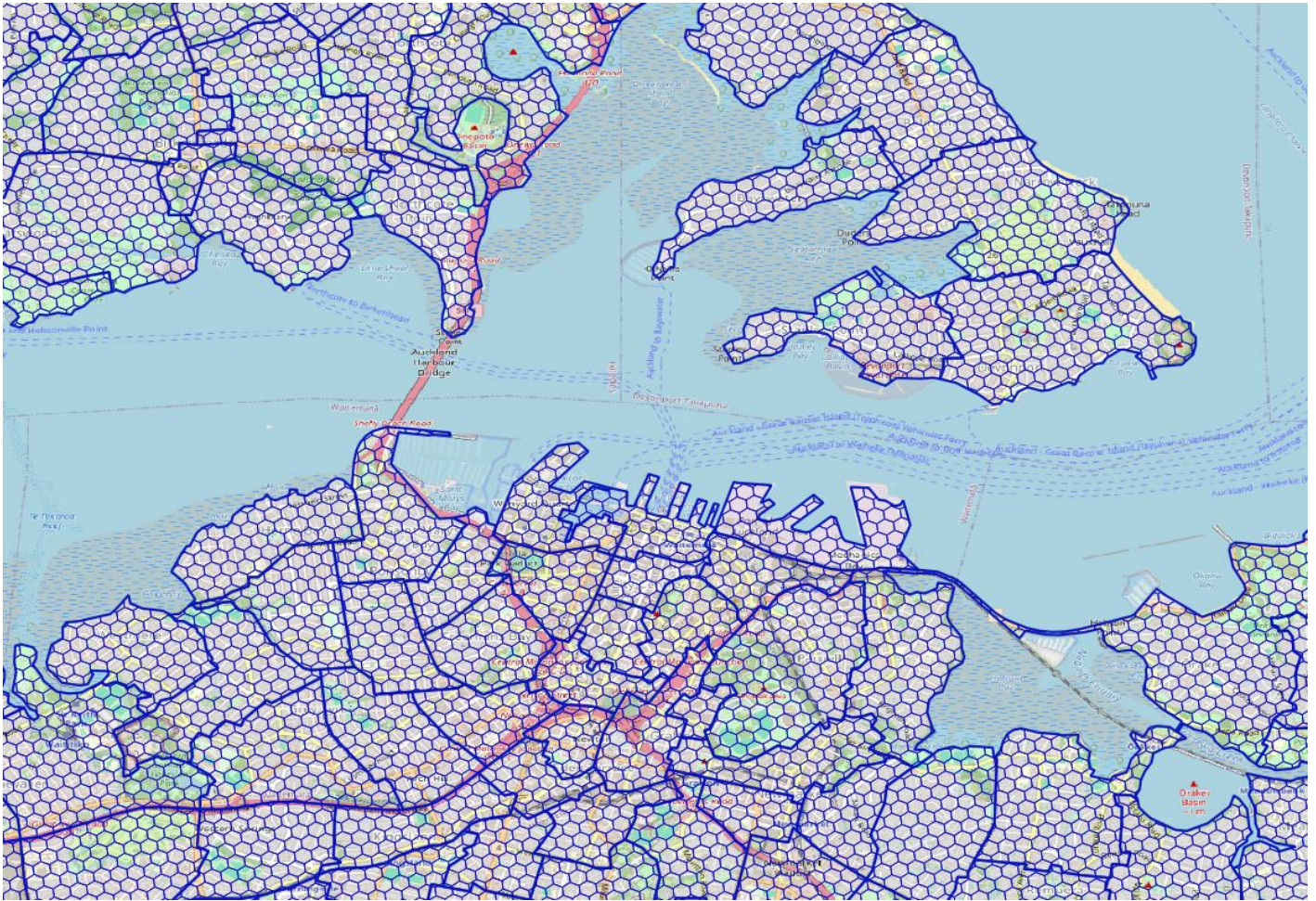


Figure 3: Spatially joined H3 hexagons and MSM Zones

7. Calculate developable area within each hexagon

- Overlay the developable-land shapefile onto the MSM-assigned hexagons.
- For each hexagon (including split hexagons), calculate the developable area and express it as a proportion of the total developable area of the corresponding MSM zone.

8. If required, apply zoning-based weighting

- Extract Unitary Plan shapefiles for more intensive development zones (e.g., Terraced Housing and Apartments, Town Centre zones).
- Overlay these intensive-development layers onto the developable-land layer and the MSM-assigned hexagons.
- Apply a user-defined weight to increase the attractiveness of more intensive development parcels within each hexagon.
- Figure 4 shows more intensive development zones (red), less intensive development zones (orange) and with MSM and H3 hexagon boundaries overlaid in blue.

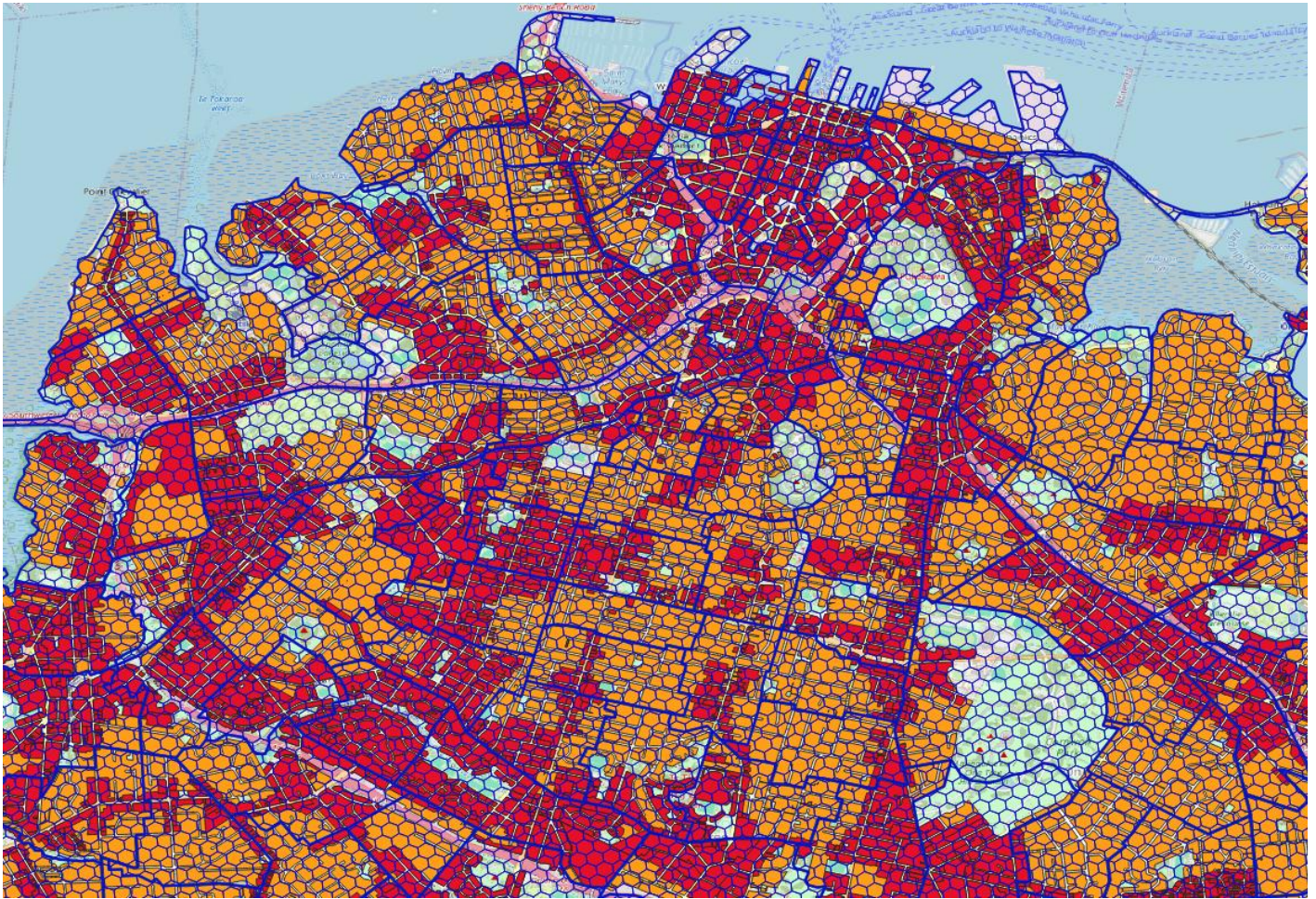


Figure 4: More intensive development overlay

9. Allocate MSM forecast demand to hexagons

- Allocate each MSM zone's forecast demand to its constituent hexagons in proportion to their share of the zone's developable area.
- Verify that the sum of allocated demand across hexagons exactly matches the original MSM forecast for each MSM zone.

10. Aggregate to the final hexagon layer

- Dissolve MSM boundaries so each hexagon has a single demand value.
- Confirm that the total demand across all hexagons equals the total MSM forecast demand across all zones.

4. Creating Allocation Factors for MSM Forecasts

A key step in the workflow is creating allocation factors to allocate demands at MSM zone level to each hexagon (or part of hexagon) located within each MSM zone. To do this, a script using Python² was developed. This script converts a combined hexagon–MSM dataset into the allocation factors required to distribute MSM forecasts to individual H{resolution level} hexagons. It cleans and validates the input data, separates developable and non-developable land, applies optional zoning-based weighting, and aggregates results for each hexagon–MSM combination. The script then

² Python Software Foundation. *Python Language Reference, version 3.x*. <https://www.python.org>

calculates weighted and unweighted proportions of developable land that sum to one for every MSM zone, ensuring full reconciliation with the original MSM forecasts. The final output is a validated, ready-to-use CSV file that enables accurate, spatially detailed allocation of MSM demand to hexagons.

Figure 5 shows the workflow followed by the script.

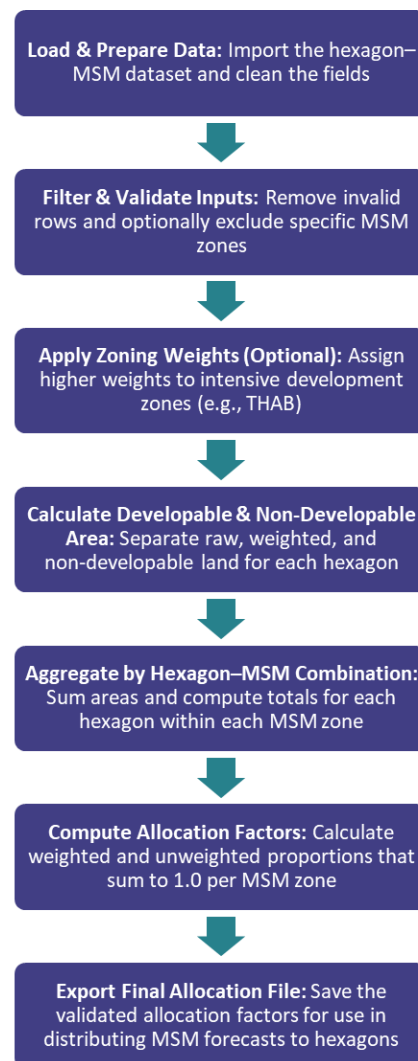


Figure 5: Workflow for Creating Hexagon Allocation Factors from MSM Inputs

4.1 Workflow

1. Load and prepare the input data

- The script loads a CSV file containing all H10 hexagons, their associated MSM zones, and the developable land area within each hexagon.
- Column names are cleaned to remove stray spaces.
- Key fields (H10, MSM, ZONE, developable area) are converted to numeric values so they can be used reliably in calculations.
- Any rows with missing or invalid MSM or H10 values are removed.

2. Optional: Exclude specific MSM zones

- The user can choose to exclude particular MSM zones from the calculation (e.g., islands, special cases, or zones outside the study area).

- Only valid MSM zones are retained for further processing.

3. Apply zoning-based weighting (optional)

- The script allows the user to apply a higher weight to land in specific Unitary Plan zones (e.g., Zone 9 = THAB).
- Zone 8 is always given a weight of 1.0; Zone 9 can be assigned any weight the user chooses.
- Two development-area measures are created:
 - **Raw developable area** (no weighting)
 - **Weighted developable area** (biasing more intensive zones)

This allows the user to test scenarios where higher-intensity zoning attracts more future activity.

4. Separate developable and non-developable land

For each hexagon:

- Land in Zones 8 or 9 is treated as developable.
- All other land is treated as non-developable.
- The script calculates:
 - Raw developable area
 - Weighted developable area
 - Non-developable area

These values are later aggregated by MSM zone.

5. Aggregate results by hexagon–MSM combination

The script groups the data by:

- H10 hexagon ID
- MSM zone ID
- The area of the hexagon within the MSM zone
- The total area of the MSM zone

For each group, it sums:

- Raw developable area
- Weighted developable area
- Non-developable area

It then calculates:

- Total area (raw)
- Total area (weighted)

These totals are used to validate that the developable-area calculations match the original input.

6. Calculate allocation proportions

For each MSM zone, the script calculates:

- The **weighted proportion** of developable land in each hexagon
- The **unweighted proportion** of developable land in each hexagon

These proportions are the **allocation factors** used to distribute MSM forecasts to hexagons.

The script also performs strict validation checks to ensure:

- All proportions within each MSM zone sum to exactly 1.0
- Weighted and unweighted totals reconcile correctly

7. Finalise and export the results

- All numeric fields are rounded for clarity.
- A diagnostic “Delta” field shows the difference between weighted and unweighted proportions.

The final dataset is saved as **MSM_H{resolution level}_Attraction_MASTER.csv**, ready for use in the allocation workflow. A description of the fields in the output csv is shown in Table 1:

Table 1: Description of Output CSV Fields

Field	Meaning
H10	Unique ID of the H10 hexagon.
MSM	MSM zone ID that the hexagon falls within.
h10_msm	Combined hexagon–MSM identifier used for grouping.
h10m_area	Area of the hexagon that lies inside the MSM zone (m ²).
MSM_Area	Total area of the MSM zone (m ²).
H_DA_raw	Total raw developable area in the hexagon (Zones 8 & 9 only).
H_DA_wt	Weighted developable area (after applying Zone 9 weight).
H_NDA	Non-developable area in the hexagon (all other zones).
Tot_HNDA_raw	Total area (raw developable + non-developable). Used for validation.
Tot_HNDA_wt	Total weighted area (weighted developable + non-developable).
MSM_Prop_Emp_wt	Weighted proportion of developable area within the MSM zone. Used for weighted allocation.
MSM_Prop_Emp_unwt	Unweighted proportion of developable area within the MSM zone. Used for baseline allocation.
Delta	Difference between weighted and unweighted proportions (diagnostic only).

Table 2 shows a sample of **MSM_H10_Attraction_MASTER.csv** for MSM zones 249 - located in the City Centre - and 266 – located in the City Centre fringe close to Kingsland.

Table 2: Sample of output CSV for MSM zones 249 and 266

H10	MSM	h10_msm	h10m_area	MSM_Area	H_DA_raw	H_DA_wt	H_NDA	Tot_HNDA_raw	Tot_HNDA_wt	MSM_Prop_Emp_wt	MSM_Prop_Emp_unwt	Delta
128196	249	128196_249	15,911	486,719	15,546	42,753	364	15,911	43,117	0.06337	0.06337	0.00000
127108	249	127108_249	15,750	486,719	14,049	38,634	1,701	15,750	40,335	0.05727	0.05727	0.00000
128010	249	128010_249	15,910	486,719	13,729	37,754	2,181	15,910	39,936	0.05596	0.05596	0.00000
124792	266	124792_266	15,914	364,950	13,440	13,441	2,474	15,914	15,914	0.02817	0.06338	0.03521
127704	249	127704_249	15,910	486,719	12,945	35,598	2,966	15,910	38,564	0.05277	0.05277	0.00000
124286	266	124286_266	15,913	364,950	12,598	34,645	3,315	15,913	37,960	0.07262	0.05941	-0.01321
124594	266	124594_266	15,913	364,950	12,221	33,608	3,692	15,913	37,300	0.07044	0.05763	-0.01282
127604	249	127604_249	15,911	486,719	11,965	32,904	3,946	15,911	36,850	0.04877	0.04877	0.00000
127311	249	127311_249	15,911	486,719	11,934	32,819	3,977	15,911	36,796	0.04865	0.04865	0.00000
127703	249	127703_249	15,476	486,719	11,728	32,253	3,747	15,476	36,000	0.04781	0.04781	0.00000
128009	249	128009_249	15,910	486,719	11,493	31,607	4,417	15,910	36,024	0.04685	0.04685	0.00000
124492	266	124492_266	15,914	364,950	11,417	11,417	4,497	15,914	15,914	0.02393	0.05384	0.02991
127602	249	127602_249	13,251	486,719	11,342	31,190	1,910	13,251	33,100	0.04623	0.04623	0.00000
128196	249	128196_249	15,911	486,719	15,546	42,753	364	15,911	43,117	0.06337	0.06337	0.00000

The *Delta* field shows how much the optional zoning-based weighting changes a hexagon’s share of developable land within its MSM zone. It is calculated as the difference between the unweighted and weighted proportions.

- A **positive Delta** means the hexagon receives a **smaller share** of demand under the weighted method (i.e., weighting shifts demand *away* from this hexagon).
- A **negative Delta** means the hexagon receives a **larger share** under the weighted method (i.e., weighting shifts demand *towards* this hexagon, typically because it contains more intensive development zoning).
- A Delta of **zero** indicates that weighting has no effect. Delta is included for transparency and diagnostic purposes only; it does not affect the final allocation unless weighted factors are selected.

5. Workflow for Allocating MSM Forecasts to H3 Hexagons

The next major step is taking MSM-level forecasts and distributes them to individual H3 hexagons using pre-calculated allocation factors. At every stage, the script checks that totals remain consistent. The result is a set of hexagon-level demand estimates that exactly match the original MSM forecasts when summed back up—ensuring accuracy, transparency, and full reconciliation.

5.1 Workflow

The workflow follows six main stages (Figure 6).

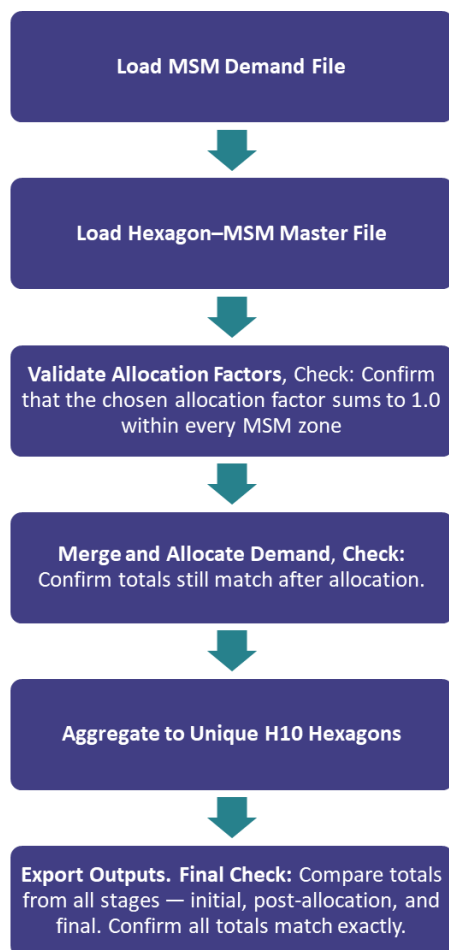


Figure 6: Allocating MSM Forecasts to H3 Hexagons

1. Load and prepare the MSM demand file

- The script loads the MSM-level demand file provided by the user.
- MSM zone IDs are cleaned and standardised so they match the format used in the allocation-factor master file.
- All demand fields are converted to numeric values.
- A diagnostic check sums all MSM-level demands before any processing begins.
 - *This establishes a baseline total that must be preserved throughout the workflow.*

2. Load the hexagon–MSM master file and align the datasets

- The script loads the **MSM_H{resolution level}_Attraction_MASTER.csv** file containing the allocation factors for each hexagon.
- Rows with invalid MSM or H10 values are removed.
- MSM zone IDs are cleaned and normalised in the same way as the demand file.
- The script identifies the MSM zones that appear in both datasets.
 - Any MSM zones that appear in one file but not the other are flagged.
 - The script stops if there are no overlapping MSM zones.
- Both datasets are filtered so they contain **only MSM zones that appear in both files**.
 - *This ensures perfect alignment before allocation begins.*

3. Validate the allocation factors

- The user chooses whether to use **weighted** or **unweighted** allocation factors.
- The script checks that the chosen allocation factor field sums to **exactly 1.0** within every MSM zone.
 - If any MSM zone does not sum to 1.0 (within a small tolerance), the script stops.
 - *This guarantees that 100% of each MSM zone's demand will be allocated to its hexagons.*

- Table 3 shows an example of the diagnostic results displaying the imported values and confirming that the allocation factors sum to 1 for each MSM zone:

Table 3: Diagnostic after Step 1

<i>metric</i>	<i>value</i>
<i>n_msm_zones_in_demand</i>	596
<i>n_msm_zones_in_master</i>	596
<i>n_msm_zones_in_both</i>	596
<i>n_msm_prop_sum_to_1</i>	596
<i>n_msm_prop_not_sum_to_1</i>	0
<i>allocation_factor_used</i>	msm_prop_emp_wt
<i>tolerance_used</i>	0.001
<i>sum_pop_26</i>	1,760,162
<i>sum_emp_26</i>	843,835
<i>sum_pop_31</i>	1,870,125
<i>sum_emp_31</i>	890,425
<i>sum_pop_41</i>	2,080,977
<i>sum_emp_41</i>	975,341
<i>sum_pop_51</i>	2,265,842
<i>sum_emp_51</i>	1,080,484
<i>sum_31_gen</i>	99,318
<i>sum_31_attr</i>	99,257
<i>sum_41_gen</i>	116,641
<i>sum_41_attr</i>	116,543

4. Merge the datasets and allocate demand

- The MSM demand file is merged with the hexagon master file using MSM zone ID.
- Any missing demand values after merging are identified and replaced with zero.
- A diagnostic check confirms that the total demand after merging still matches the original MSM totals.
- The script multiplies each hexagon's allocation factor by the MSM-level demand to calculate the demand allocated to each hexagon.
- A second diagnostic check confirms that the total demand **after allocation** still matches the original MSM totals.
 - *This ensures that no demand is lost or double-counted during allocation.*

5. Aggregate results to unique H10 hexagons

- The script groups the allocated results by H10 hexagon ID.

- All allocated demand values for each hexagon are summed.
- A diagnostic check confirms that the total demand across all hexagons still matches the original MSM totals.
 - *This ensures that splitting hexagons across MSM boundaries does not distort totals.*

6. Final consistency check and export

- A final three-way comparison is performed:
 - Initial MSM totals
 - Post-allocation totals
 - Final hexagon totals
- If all three totals match, the script confirms success – Table 4 shows an example of the diagnostic output.

Table 4: Final Step Allocation Diagnostics

<i>Demand Field</i>	<i>Step 1 (Initial)</i>	<i>Step 2 (After Applying Prop)</i>	<i>Step 3 (Final)</i>
<i>31_attr</i>	99,257	99,257	99,257
<i>31_gen</i>	99,318	99,318	99,318
<i>41_attr</i>	116,543	116,543	116,543
<i>41_gen</i>	116,641	116,641	116,641
<i>emp_26</i>	843,835	843,835	843,835
<i>emp_31</i>	890,425	890,425	890,425
<i>emp_41</i>	975,341	975,341	975,341
<i>emp_51</i>	1,080,484	1,080,484	1,080,484
<i>pop_26</i>	1,760,162	1,760,162	1,760,162
<i>pop_31</i>	1,870,125	1,870,125	1,870,125
<i>pop_41</i>	2,080,977	2,080,977	2,080,977
<i>pop_51</i>	2,265,842	2,265,842	2,265,842

- Two output files are created:
 1. **Hexagon-level allocation file** (before aggregation)
 2. **Final H10 aggregated file** (one row per hexagon)

These outputs are ready for mapping, analysis, and integration into the wider modelling workflow.

6. Creating GTFS Feeds from MSM Public Transport Inputs

To support high-resolution accessibility analysis on the H3 hexagon grid, MSM's public transport network inputs are converted into General Transit Feed Specification (GTFS) format. GTFS provides a standardised, route- and trip-level representation of public transport services that can be used directly by tools such as *r5r* for multimodal travel time and accessibility modelling. This section summarises the workflow used to generate GTFS feeds from MSM model outputs.

6.1 Overview

MSM produces public transport network information as a set of shapefiles for each scenario and time period, including *tlines*, *tsegs*, *links*, *nodes*, and *turns*. These files contain route alignments, segment distances, stop locations, and service patterns. A structured Python workflow transforms these MSM shapefiles into a complete GTFS feed by extracting, cleaning, and recombining the required elements into the standard GTFS tables.

6.2 Workflow for GTFS Feed Creation

The GTFS creation process follows a consistent sequence across all MSM scenarios:

1. **Prepare MSM shapefiles**

MSM shapefiles are loaded and renamed to a consistent scenario-based naming convention. Segment distances are added to the *tsegs* layer to support accurate stop-to-stop travel times.

2. **Export shapefiles to CSV**

All MSM shapefiles are converted to CSV format to enable downstream processing. These CSVs form the base inputs for constructing GTFS tables.

3. **Create core GTFS tables**

A series of Python scripts generate the required GTFS components:

- *agency.txt* — defines the public transport operator.
- *routes.txt* — created from *tlines*, representing each service pattern.
- *trips.txt* — links routes to individual trips and service patterns.
- *stop_times.txt* — constructed from *tsegs* and segment distances, providing ordered stop sequences and travel times.
- *stops.txt* — generated from *tsegs* to define stop locations and identifiers.
- *shapes.txt* — derived from route geometries to represent the physical alignment of each service.
- *calendar.txt* and *frequencies.txt* — define service days and headways for each trip. Because MSM public transport inputs are frequency-based rather than fully timetabled, the GTFS feed requires manual adjustment of rail service start times to ensure realistic headways at stations and to avoid multiple services being scheduled to arrive or depart simultaneously.

4. **Assemble and package the GTFS feed**

All GTFS tables are written to text files and packaged into a compressed GTFS feed (.zip) for each MSM scenario. The resulting feed is fully compliant with GTFS standards and can be used directly in *r5r* or other accessibility modelling tools.

6.3 Application

The GTFS feeds produced through this workflow allow MSM's public transport networks to be analysed at the same spatial resolution as the H3 hexagon demand allocation. This enables consistent, high-resolution accessibility modelling across scenarios, years, and policy settings, supporting more detailed and spatially faithful assessment of public transport performance.

Appendix A: Spatial workflow to calculate factors for allocating MSM forecasts to hexagons

Steps	QGIS Task	Input Shapefile	Output Shape File	Method
1	Load H10 shapefile	h10_base_25Oct25.shp		
2	Load MSM shapefile	MSM_FINAL_10Aug25.shp		
3	Intersect H10 with MSM	h10 and MSM shapefiles	h10_msm_master_19dec25.shp	Input layer: h10_base_25Oct25.shp (this will be the geometry used for output). Overlay layer: MSM_FINAL_10Aug25.shp. Set the output file location and format (e.g., ESRI Shapefile)
4	Create a new field " h10_msm ", and insert area field " h10m_area "	h10_msm_master_19dec25.shp	h10_msm_master_19dec25.shp	create a new field concatenating H10 and MSM zones: to_string("H10") '-' to_string("MSM") Create area field "h10m_area" and insert area: area(\$geometry)
5	Load shapefile with development areas from UP layers	up_emp_valid_weight_19dec25.shp	up_emp_valid_weight_19dec25.shp	
6	Create shapefile retaining the full coverage of h10_msm_master_19dec25.shp with geometries in h10_msm to be split along UP_EMP boundaries where they intersect, while preserving all original zones.	up_emp_valid_weight_21dec25.shp h10_msm_master_19dec25.shp	h10_dev_union_19dec25.shp	Open Processing Toolbox → Vector Overlay → Union Input layer: h10_msm_master_19dec25 Overlay layer: up_emp_valid_weight_21dec25
7	Save the union temporary scratch file as a shapefile	union	h10_dev_union_19dec25.shp	
8	Clean h10_dev_union_14Jul25.shp			Open your Attribute Table Click the Select by Expression icon (looks like an epsilon Σ) Use this expression: "H10" = 0 OR "H10" IS NULL

9	Create new field "tot_hdna" to store the area of each record	h10_dev_union_19dec25.shp	h10_dev_union_19dec25.shp	Toggle Editing Mode Open Attribute Table Open Field Calculator Enable "Create a New Field" Fill In Field Details - Field name: no-spaces name (tot_hdna) - Field type: Decimal number (real) for areas - Length: 20, and Precision: 2 for numeric field Open Field Calculator Select "Update Existing Field" - Choose "tot_hdna" from the drop-down list (don't create a new field again). - Enter the Area Expression: area(\$geometry)
10	Export to csv file for input to python to create the factors to allocate land use and demand at MSM (MSM) zone level to hexagons	h10_dev_union_19dec25.shp	h10_dev_union_19dec25.csv	
Steps	Python Task	Python script	Input file	Output file
1	Calculate factors to allocate development / demand at MSM zone level to hexagons.	calculate H10 to MSM Development Proportions Allocation 19dec25.py	h10_dev_union_21dec25.csv	MSM_H10_Attraction_MASTER.csv
2	Allocate MSM level demands of land uses to hexagons	allocate demand and land use at MSM zone level to H10 hexagons 21dec25.py		
3	Create shapefile for plotting in R (2D or Rayshader)			